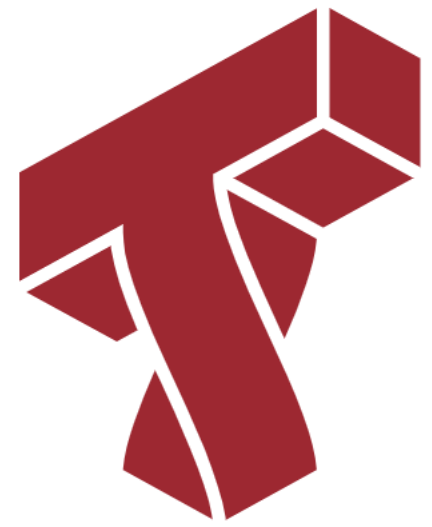




CS 5785 / ECE 5414 / ORIE 5750

Applied Machine Learning

Fall 2026

Kyra Gan

TAs: Adnan Armouti, Jindan Li
Jiamin Xu, Diyang Li

M/W 7:30 pm – 8:45pm Bloomberg Center, Room 131

CS 5785/ECE 5414/ORIE 5750 Applied Machine Learning

Lecture 1

Introduction to Machine Learning

Course Logistics

CS5785/ORIE5750/ECE5414 > Syllabus

6d View as Student



Fall 2026

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Course Syllabus

Jump to Today

Edit

Instructor: Prof. Kyra Gan (<https://kyra-gan.github.io/>)

Meeting Room: Auditorium (Bloomberg Center, Room 131)

August 2026						
<						>
26	27	28	29	30	31	1
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23	24	25	26	27	28	29

<https://canvas.cornell.edu/courses/88328>

science, information theory, optimization, optics, and ecology. In short, machine learning is a broad field. In this course, we will cover the breadth, there's a common thread: machine learning is concerned with developing algorithms (and an evaluation metric) as they acquire more experience.

This course aims to equip students with the fundamental mathematical skills needed to understand, interpret, and adapt modern machine learning models. Rather than focusing on engineering models for optimal performance, we will emphasize developing a rigorous mathematical understanding of how these models work. This foundation will enable students to independently learn and modify cutting-edge models after completing the course. From an applied perspective, we will examine the core assumptions behind each machine learning approach and learn to evaluate whether a model is appropriate for specific real-world applications. The curriculum will cover essential concepts in supervised and unsupervised learning, with the latter portion dedicated to understanding classical and contemporary deep learning architectures.

Course Objectives/Learning Outcomes:

By the end of this class, you will be able to:

- Articulate the fundamental problems addressed by machine learning and their exemplar applications, and be conversant with established tools of the trade.
- Describe and derive linear models and neural networks for arbitrary applications.

Teaching Staff

Instructor: **Prof. Kyra Gan** (<https://kyra-gan.github.io/>)

- Assistant Professor in ORIE
 - My fourth year here
- My research:
 - Causality, sequential decision making
 - Applications in healthcare (heart failure, single cell...)

Teaching Staff

Instructor: **Prof. Kyra Gan** (<https://kyra-gan.github.io/>)

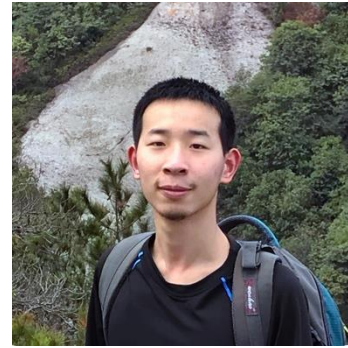
TAs:



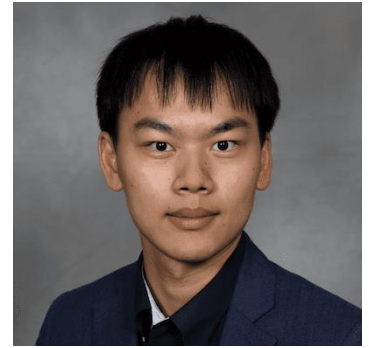
Adnan Armouti



Jindan Li



Diyang Li



Jiamin Xu

Full Time

Part Time

Graders:

Jiwon Jeong (jj835), Zirui Han (zh524), Fengze Cai (fc389),

Ruixiang Lu (rl974), Kaiwen Zhoing (kz388)

You?

Course Objectives: By the end of the course you will be able to ...

- Understand the fundamentals of how machine learning algorithms work
 - The what, the why, and the how (the math)
- Propose a workflow for a real-world problem
 - How to collect/preprocess data; which algorithms to use; how to evaluate model performance
- Assess whether an ML algorithm is suitable for the proposed real-world problem
 - What assumptions do the ML require? Do they make sense in the problem setup?
 - How to improve the performance of existing algorithms

Course Objectives: By the end of the course you will be able to ...

- Build your own ML tools
- Work on real-world problems
- Course projects and homeworks

Course Contents

- Lectures: foundations of ML algorithm in math
 - Understand the principles behind common ML methods: the required assumptions, when do they work?
- Coding:
 - In-class lab/review sessions
 - Homework & Office hours
 - python + jupyter notebook
 - Packages: matplotlib, numpy, pandas, sklearn, pytorch
 - Best compromise of powerful scripting language and most common data wrangling and ML tools

Topics

- First half of the semester:
 - Basic supervised learning:
 - Regression + gradient descent, classification + maximum likelihood estimation, regularization, generative vs discriminative models + expectation maximization algorithms
 - Unsupervised learning:
 - Clustering, Density estimation (nonparametric estimation), dimensionality reduction
- Second half:
 - Deep learning: multilayer perceptron, CNN
 - Advanced topics: diffusion, RL

Tentative Class Schedule

Date	No	Topic	Assignments
8/24, M	1	Introduction/overview of the class	
8/26, W	review	In class review: probability + linear algebra	
8/31, M	review	In-class review: calculus + coding autograd	
9/2, W	2	Supervised learning + linear regression	HW0 Due (9/2, Wed)
9/7, M		labor day, no class	
9/9, W	3	empirical risk + gradient descent	
9/14, M	4	gradient descent + logistic regression	
9/16, W	5	Classification and Maximum likelihood estimator	HW1 Due (9/16, Wed)
9/21, M	6	classification evaluation, choice of divergence, and data underspecification	
9/23, W	7	Regularization, Model Selection, and Generative Model I	
9/28, M	8	Generative model and Text Classification	
9/30, W	9	Naive Bayes and GMM	HW2 Due (9/30, Wed)
10/5, M	10	unsupervised learning and k-means	
10/7, W	11	GMM and Expectation Maximization Algorithm	
10/12, M		No Class, Fall Break, Indigenous Peoples' Day	
10/14, W	12	EM, nonparametric estimation, Density estimation, kernels	
10/19, M	13	KNN	HW3 Due (10/19, Mon)
10/21, W	14	Dimensionality reduction	10/23: Project one paragraph description
10/26, M	15	Guest Lecture: sensor/vision	
10/28, W	16	PCA, Decision trees, random forest	
11/2, M		No class, midterm due	Midterm project: semisupervised learning c
11/4, W	17	Distributional shifts and Neural networks	
11/9, M	18	NN II	
11/11, W	19	Learning multilayer preceptrons, regularization, and other tricks	
11/16, M	20	dropout and CNNs	
11/18, W	21	CNN II	HW4 Due (11/18, Wed)
11/23, M	22	Guest lecture: attention, diffusion	
11/25, W		Thanks giving break	(11/26) Revised one-pager description
11/30, M	23	RL I	
12/2, W	24	RL II	
12/7, M	25	Guest Lecture: TBD	HW5 Due (12/7, Mon)
12/17: final project due			

How is this section different?

- For the most part, it really isn't
 - Both sections are an intro to ML with a complete practitioner's toolkit, potentially with a different pace
- Midterm:
 - Check the syllabus of the other session
- Homework:
 - This class: additional canvas quiz component, multiple choices
- Might cover slightly different special topic classes:
 - This class: reinforcement learning, LLM...

Logistics: office hours

- We want to schedule our office hours so to offer maximum availability to you.
- Please fill out the office hour survey on Canvas **by next Monday**
 - Under Assignments
- First-week office hours:
 - Kyra: by appointment
 - TA: TBA

Communications

- Office hours: Best place for questions outside of class
- Canvas Ed Discussions: Second-best place online
 - Click the “Ed Discussions” tab on Canvas
- Watch out for Canvas Announcements
 - Enable notification on Canvas settings
 - Click the “Announcement” tab in Canvas
- Email: cc everyone (**including your TAs**) to ensure a response

Three In-class Review Sessions

Probability/Linear Algebra & Calculus/Gradient Descent review session:

- This Wednesday (8/26) and next Monday (8/31) in class
- Zoom links can be found on Canvas under the Zoom tab
- Recording will be available

Three In-class Review Sessions

Coding review session:

- Next Monday (**Aug 31, 2026** 2:55 - 4:10 PM), Bloomberg 131
- Will be recorded and uploaded to Canvas
- Can attend via Zoom (Zoom link to be announced later on Canvas)

Remote Attendance

To optimize your learning experience:

- **In-person attendance is required unless you are encountering visa delays or related issues**

Visa delays after Week 2? Please email me to discuss your situation. I will provide a Zoom option for students with ongoing, verified delays.

In-class quizzes &

HW questions covering in-class quizzes



In-class Participation Quiz

- Full points as long as you answer
- Scan the QR code or go with the link
- Log in with your Cornell email address
- It will prompt for SSO login

Who are your TAs for this class?



Adnan Armouti

0

Jindan Li

0

Diyang Li

0

Jiamin Xu

0

All of the above

0

Logistics: grade breakdown

- Homework assignments (40%)
 - 5 Homeworks + Homework 0
 - Canvas Quiz on lecture material + Gradescope submission
- Midterm Group Kaggle Competition (25%)
 - ~One week to complete
- Final group project (25%)
- Participation (10%)
 - Attendance (poll everywhere) and class survey participation (60%)
 - A+: be active in class (40%)

Designed so everyone can and should do well

Logistics homework late policy

- **6** slip days in total
 - May be used without penalty
 - A maximum of **2** slip days for any single assignment.
- Beyond that, any late submission is subject to an additive **20%** penalty for each late day

Logistics: Homework 00

- 2 Surveys:
 - One on your background
 - One on office hours

Logistics: Homework 0

- The "zeroth" homework assignment will be posted under the Gradescope page on Canvas.
- ~One week from today to complete
- Setting up your Python environment
- Installing packages
- Review some basic questions
- Familiarizing yourself with our submission system
- **Not graded:** you will receive full points as long as you submit something.
- If you have issues, come to the **TA** office hours

Logistics: collaboration

- You are expected to complete each homework assignment independently
- Midterm and final project
 - You can create your own group of sizes up to 3
- If you're looking for a partner, try **Canvas Ed Discussions**

When are the three in-class review sessions?

8/26 (wed) 11:40 - 12:55 pm

0

8/26 (wed) 7:30 - 8:45 pm

0

8/31 (mon) 7:30 - 8:45 pm

0

9/2 (wed) 7:30 - 8:45 pm

0

8/31 (mon) 2:55 - 4:10 pm

0

Classroom Engagement

- This is an interactive class
- We like questions
 - No question is silly/wrong/embarrassing
 - If you have a question, odds are that others have it too

Please participate



Classroom Engagement

- Keep your smart devices shut
- No phones or laptops open except for the following:
 - To answer polls
 - To view lecture slides
 - To take notes

Please do not use AI **in classroom** to understand the course material

- You will likely get lost for the rest of the class
- You should practice communication with humans

Resources

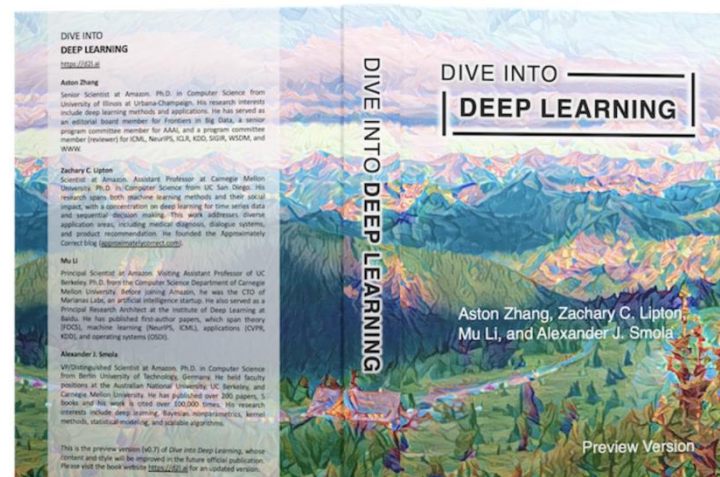
Recommended: (available for free online)

- Dive into Deep Learning

(web version free online at <https://d2l.ai/>)

(source code free at <https://github.com/d2l-ai/d2l-en>)

- Machine Learning by Tom Mitchell
- Machine Learning—A Probabilistic Perspective by Kevin Murphy
- Pattern Recognition and Machine Learning by Christopher Bishop



Breakthroughs with ML/Deep Learning (2016)

www.technnewsworld.com/story/84013.html

cs.stanford.edu/people/karpathy/deepimagesent/

40 maps that explain: Amazon Web Services, Primers | Math n Proc, deeplearning.net/tut, Deep Learning Tutor, deep learning, PHILIPS - Golden Ears, Language Technology, MyIDCare - Dashbo, Other bookmarks

man in black shirt is playing guitar.

construction worker in orange safety vest is working on road.

two young girls are playing with lego toy.

boy is doing backflip on wakeboard.

girl in pink dress is jumping in air.

black and white dog jumps over bar.

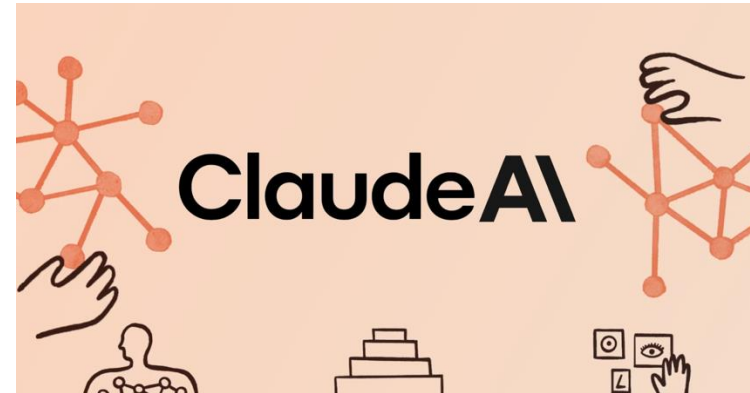
young girl in pink shirt is swinging on swing.

man in blue wetsuit is surfing on wave.

in chess, compared to 2016 ICCC world-champion program *Stockfish*. **d** Performance of *AlphaZero* in shogi, compared to 2017 CSA world-champion program *Elmo*. **c** Performance of *AlphaZero* in Go, compared to *AlphaGo Lee* and *AlphaGo Zero* (20 block / 3 day) (29).

And now...

Large Language Models



And now... Large Language Models

ChatGPT,
what animal is
this?

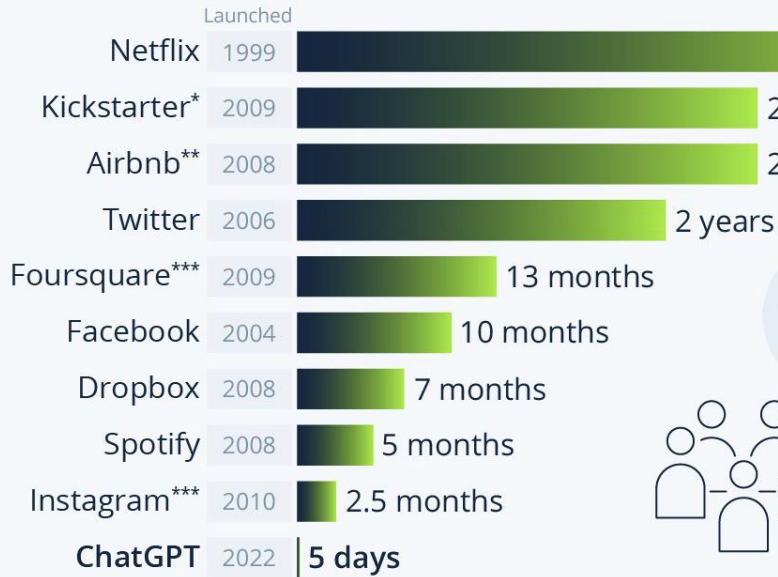


Those are **humpback whales**. The long, pale/white pectoral fins visible under the water are a strong clue, along with the small dorsal fin sitting on a noticeable hump.

Looks like there are **two of them traveling together**.

ChatGPT Sprints to One Million Users

Time it took for selected online services to reach one million users



* one million backers ** one million nights booked *** c

Source: Company announcements via Business Insider/Linl



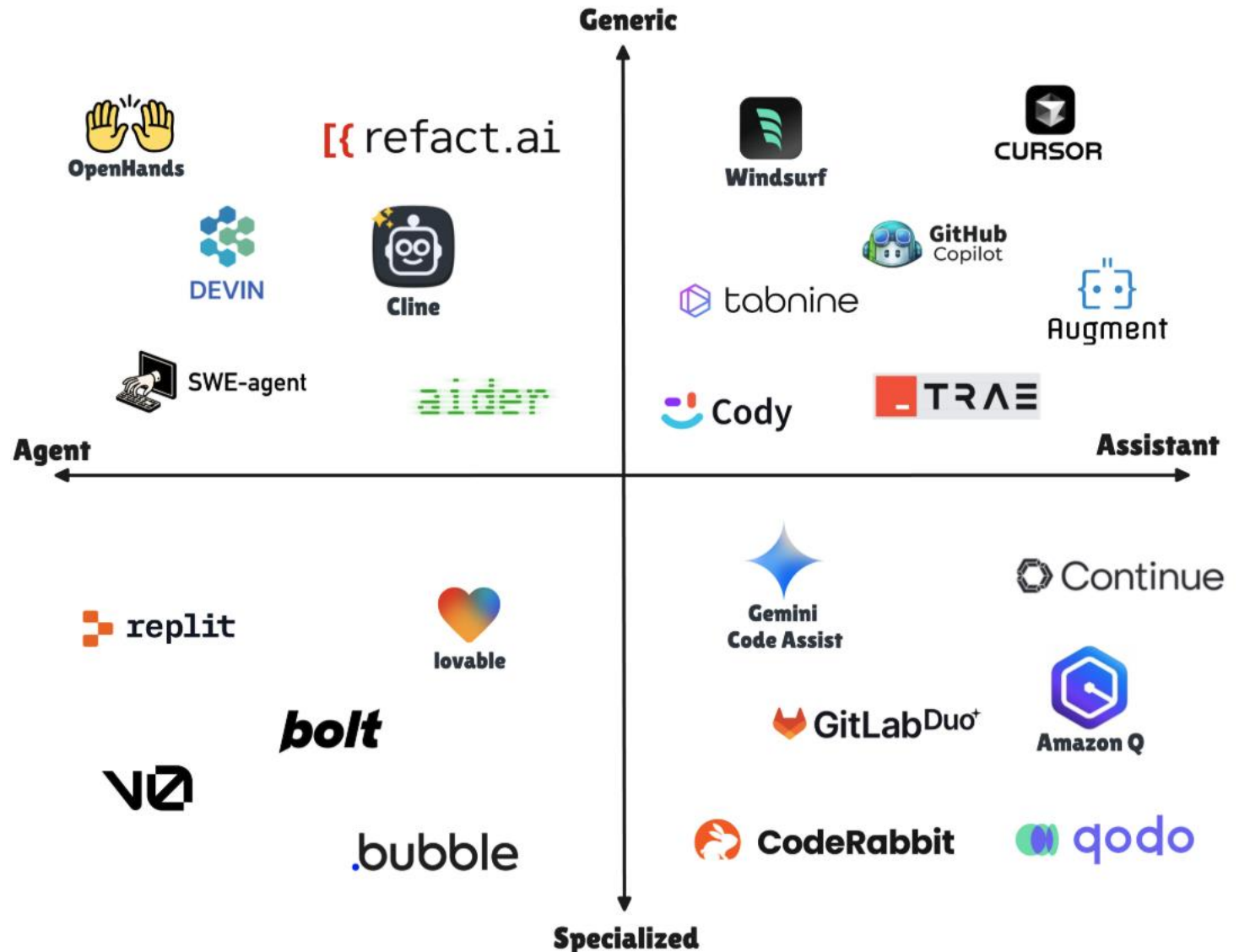
Super Products Time Spent by Users to Reach 100 Million Users



Coding Agents

AI Coding Assistants Landscape

<https://GenerativeProgrammer.com> 03/2025



Biology and Beyond

AlphaFold 3

A game-changing tool



However, Many Problems Remain Difficult...

- GenAI does not solve all problems
 - State-of-the-art models still hallucinates (although they are getting better over time)
 - They are not interpretable, and still learns association rather than causation
- We might not have the right type of data to answer some of those questions
 - e.g., which patients will develop resistance to cancer therapies?
 - **Data types and representation are important!**

Use GenAI in Assignments

- Everyone has a different background, and use AI differently
- **You should be able to develop critical thinking**
- You can ask AI to explain concepts, but you need to be aware that they make mistakes
- You should think through whether the answer is logical, and internalize the material rather than following what AI is providing
- You can ask AI to help you code, but you need to understand the logic of the code. Otherwise, you won't be able to debug, and it becomes very hard to verify/fix your code

When you use GenAI as a reference

- Reference it and explain how you used it
- If you don't understand why AI is doing something
 - It's a signal that you should review the fundamentals and
→ go to office hours
- I recommend not using AI if you are **not** comfortable with it's output
 - We need to learn how to walk before we run
- Be careful with agentic AI use

Why does ChatGPT hallucinate?

- ChatGPT is designed for tasks relating to natural language processing
- It is a generative model
 - The content that it generates can be false
- It will **fail** if you use it for
 - Self-driving cars
 - Treatment recommendation
 - Reasoning about causal relations
 - Statistical inference
 - ...

How is ChatGPT trained?

Unsupervised Learning

- A dataset without labels
- Objective: learn something interesting about the structure of the data
- Clusters hidden in the dataset.
- Outliers: particularly unusual and/or interesting datapoints.
- Useful signal hidden in noise, e.g., human speech over a noisy phone.
- ChatGPT: Pre-training
 - Next-token prediction (aka, self-supervised learning)

Supervised Learning

- The most common approach to machine learning
- First, we collect a dataset of labeled training examples
- We train a model to output accurate predictions on this dataset
- When the model sees new, similar data, it will also be accurate
- Chat GPT: Fine Tuning

Reinforcement Learning with Human Feedback

- Where do we obtain the training labels?
 - Manually labeling the data
 - Expensive

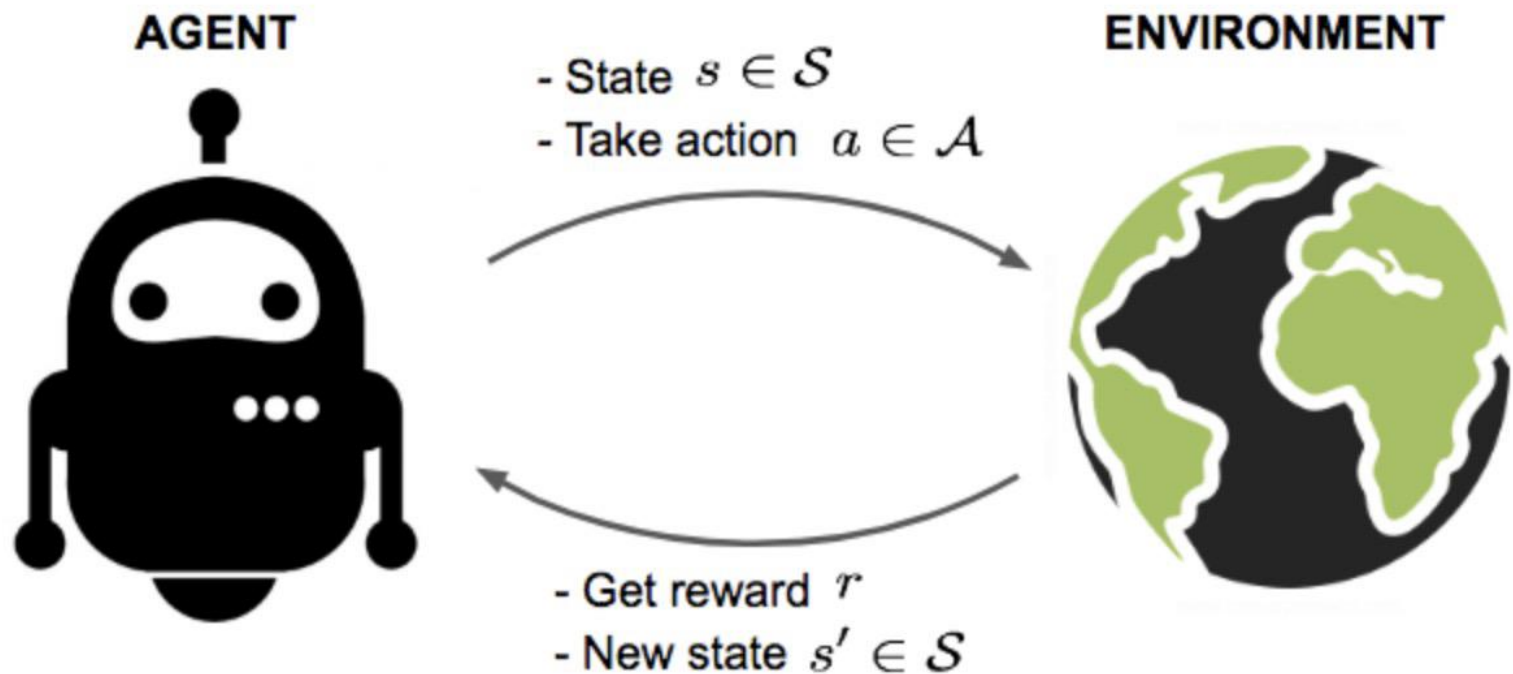
How is ChatGPT Trained?

- Unsupervised/self-supervised learning
 - Pre-training
- Supervised learning
 - Fine tuning
- RL with human feedback
 - Update Model using RL with Manually labeled preference data

The performance of ChatGPT is limited by the dataset it was trained on, and the amount of labels that are available

Reinforcement Learning

- An agent is interacting with the environment over time. We teach it good behavior by providing it with rewards.



Reinforcement Learning

- Applications of RL include:
 - Creating agents that play games, e.g., Chess or AlphaGo.
 - Controlling the cooling systems of datacenters to use energy more efficiently.
 - Digital health interventions.



Learning Objectives

1. Demystify supervised and unsupervised learning.
2. Cast real-world tasks as formal ML problems.
3. Identify and apply appropriate ML methods.
4. Master deep learning basics.
5. Critically analyze the promise and perils of ML in context.
6. Understand RL/LLM

AI and Deep Learning

- Machine learning is closely related to these two fields.
- AI is about building machines that exhibit intelligence.
- ML enables machines to learn from experience, a useful tool for AI.
- Deep learning focuses on a family of learning algorithms loosely inspired by the brain.

